# Cohort canonical evaluation — 29 agents

idea-01-lateral-attribution · 130 held-out segments · V0 yaw 0.01456 rad/s, CTE 147.44 m · generated 2026-05-31 20:34

## Executive summary

AGENTS (OK / TOTAL)

**28 / 29**

EVAL POOL

**130 segs**

V0 YAW RMSE

**0.01456**

V0 CTE RMSE

**147.44 m**

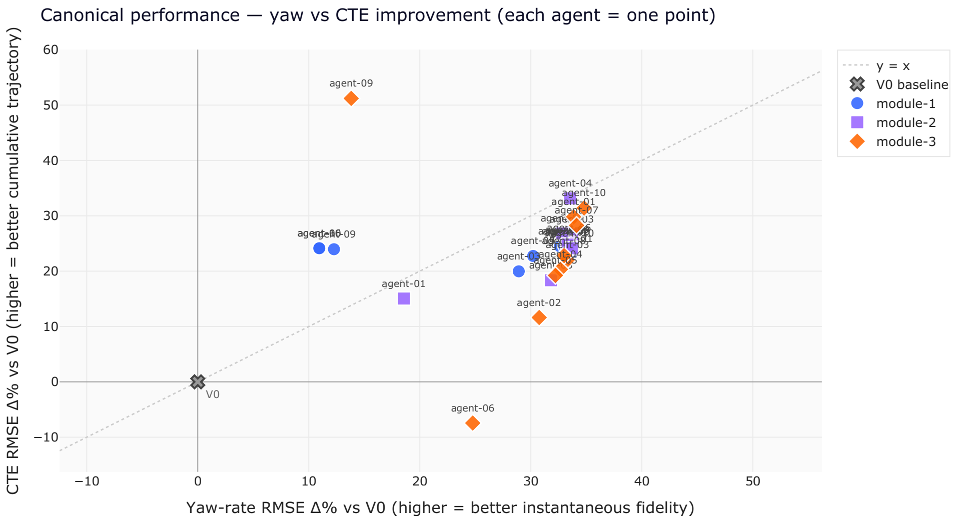
WALL TIME

**4.63s**

Best yaw: `m3-agent-10` **+34.8%** · Best CTE: `m3-agent-09` **+51.2%**
**Winning both KPIs ≥ +30%** (2 agents): `m3-agent-10`, `m2-agent-04`

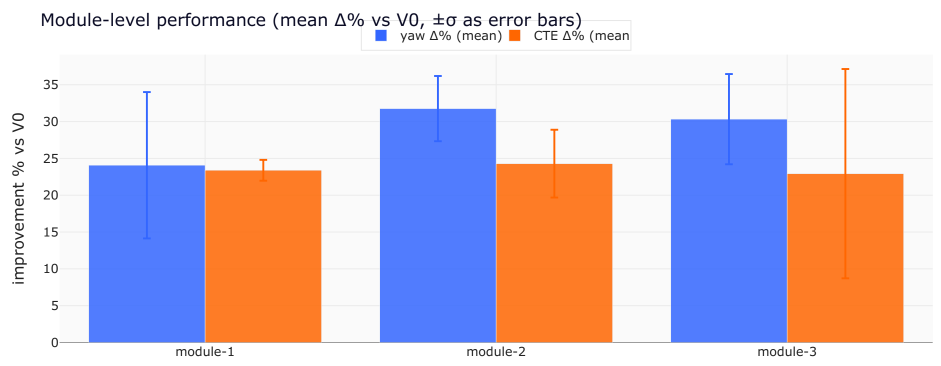
## 1. Headline scatter — yaw vs CTE improvement

Each agent is one point. **Top-right is the goal** — both KPIs improve over V0. The dashed diagonal is y = x: above it, an agent improved CTE more than yaw (good trajectory integration); below, the opposite. The V0 baseline sits at the origin.




# 2. Module-level aggregation

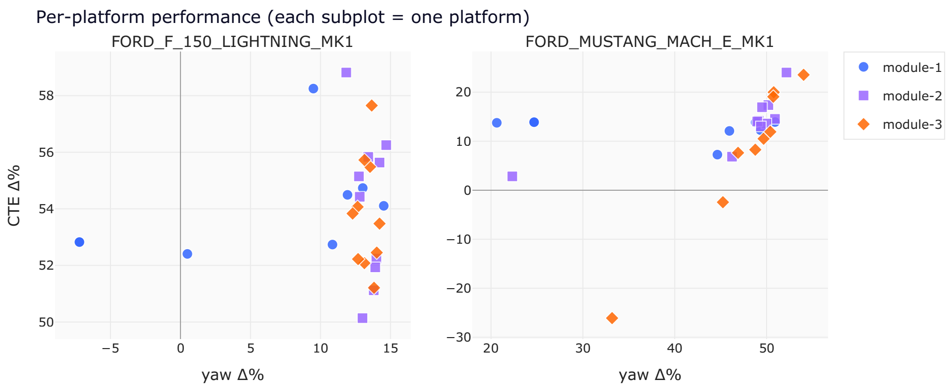
Performance grouped by which module shipped the agent. Mean Δ% over only the agents that reconstructed successfully; failures shown separately.




| family | n ok / total | yaw Δ% (mean ± σ) | CTE Δ% (m |
|---|---|---|---|
| module-1 | 8/9 | +24.1% ± 9.9% | +23.4% ± |
| module-2 | 10/10 | +31.8% ± 4.4% | +24.3% ± |
| module-3 | 10/10 | +30.3% ± 6.1% | +22.9% ± |

## 3. Per-platform breakdown

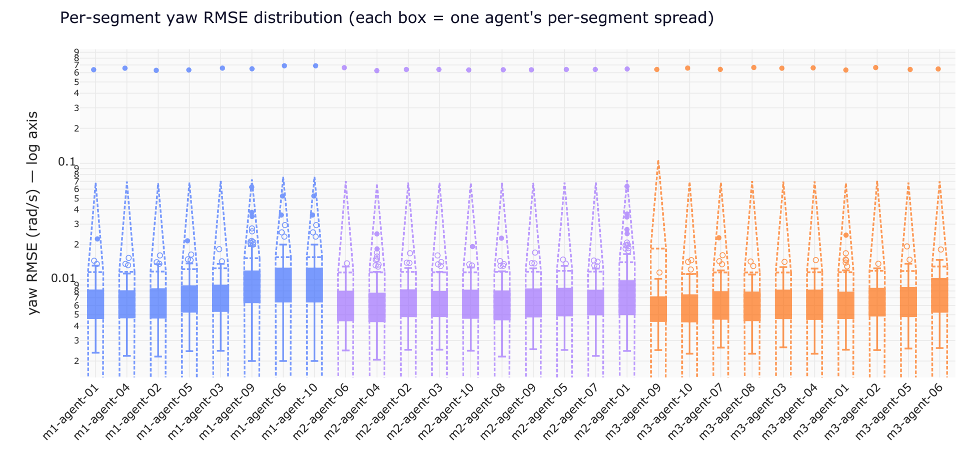
Same cohort, but split by the vehicle platform the agent was evaluated on. Agents that win one platform but lose another have a platform-specific calibration; agents whose dots cluster together generalise.




| platform | agents | yaw Δ% (mean ± σ) |
|---|---|---|
| FORD_F_150_LIGHTNING_MK1 | 28 | +11.2% ± 5.7% |
| FORD_MUSTANG_MACH_E_MK1 | 27 | +44.7% ± 9.8% |

## 4. Per-segment yaw RMSE distribution

Each box summarises one agent's per-segment yaw RMSE — pooled RMSE can hide that an agent does great on most segments but pathologically badly on a few. The y-axis is logarithmic to compress outliers.




## 5. Per-agent canonical scorecard

The full table. Failed reconstructions are surfaced with their reason — no fake zeros.

| agent | family | status | yaw V0 | yaw |
|---|---|---|---|---|
| m1-agent-01 | module-1 | ok | 0.014563 | 0.00 |
| m1-agent-02 | module-1 | ok | 0.014563 | 0.00 |
| m1-agent-03 | module-1 | ok | 0.014563 | 0.01 |
| | module-1 | ok | 0.014563 | 0.00 |

| agent | family | status | yaw V0 | yaw |
|---|---|---|---|---|
| m1-agent-04 | | | | |
| m1-agent-05 | module-1 | ok | 0.014563 | 0.01 |
| m1-agent-06 | module-1 | ok | 0.014563 | 0.01 |
| m1-agent-07 | module-1 | import_failed | — | |
| m1-agent-09 | module-1 | ok | 0.014563 | 0.01 |
| m1-agent-10 | module-1 | ok | 0.014563 | 0.01 |
| m2-agent-01 | module-2 | ok | 0.014563 | 0.01 |
| m2-agent-02 | module-2 | ok | 0.014563 | 0.00 |
| m2-agent-03 | module-2 | ok | 0.014563 | 0.00 |
| m2-agent-04 | module-2 | ok | 0.014563 | 0.00 |
| m2-agent-05 | module-2 | ok | 0.014563 | 0.00 |
| m2-agent-06 | module-2 | ok | 0.014563 | 0.00 |
| | module-2 | ok | 0.014563 | 0.00 |

| agent | family | status | yaw V0 | yaw |
|---|---|---|---|---|
| m2-agent-07 | | | | |
| m2-agent-08 | module-2 | ok | 0.014563 | 0.00 |
| m2-agent-09 | module-2 | ok | 0.014563 | 0.00 |
| m2-agent-10 | module-2 | ok | 0.014563 | 0.00 |
| m3-agent-01 | module-3 | ok | 0.014563 | 0.00 |
| m3-agent-02 | module-3 | ok | 0.014563 | 0.01 |
| m3-agent-03 | module-3 | ok | 0.014563 | 0.00 |
| m3-agent-04 | module-3 | ok | 0.014563 | 0.00 |
| m3-agent-05 | module-3 | ok | 0.014563 | 0.00 |
| m3-agent-06 | module-3 | ok | 0.014563 | 0.01 |
| m3-agent-07 | module-3 | ok | 0.014563 | 0.00 |
| m3-agent-08 | module-3 | ok | 0.014563 | 0.00 |
| | module-3 | ok | 0.014563 | 0.01 |

| agent | family | status | yaw V0 | yaw |
| --- | --- | --- | --- | --- |
| m3-agent-09 | | | | |
| m3-agent-10 | module-3 | ok | 0.014563 | 0.00 |

## 6. Calibration cards — fitted coefficients

Where the cohort converges on similar physics, and where it forks. Each tile is one agent's `coeffs.json` as shipped.

### m1-agent-02


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "s_scale": 0.94203,
    "delta_offset_rad": 0.00102,
    "K_us": 0.00304,
    "tau_yaw_s": 0.04592
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "s_scale": 1.21716,
    "delta_offset_rad": -5e-05,
    "K_us": 0.00304,
    "tau_yaw_s": 0.05262
  },
```

```
  "TESLA_MODEL_3": {
    "L": 2.875,
    "s_scale": 1.0,
    "delta_offset_rad": 0.0,
    "K_us": 0.0025,
    "tau_yaw_s": 0.05
  },
  "_method": "Fit on 60% of segments per platform; model: yr_ss = 
  "_tesla_note": "TESLA_MODEL_3 has no measured yaw-rate truth; co
}
```

## m1-agent-04


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "a": 0.9127536047016024,
    "b": 0.0007692876565060736,
    "delta_off": 0.0012208376981866274,
    "tau": 0.06378078127896805
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "a": 1.1599450659882828,
    "b": 0.0008023134078587542,
    "delta_off": 1.3682104928342577e-05,
    "tau": 0.06877523527457137
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "a": 1.0,
    "b": 0.0,
    "delta_off": 0.0,
    "tau": 0.0
  }
}
```

m2-agent-01


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "K_us": 0.0045433459530302405,
    "delta0": 0.0013939755442043738,
    "tau": 0.059696812148648346
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "K_us": 0.0008613349618482884,
    "delta0": -2.4463406949899732e-05,
    "tau": 0.058058773969275335
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K_us": 0.0007,
    "delta0": 0.0,
    "tau": 0.08
  }
}
```

## m2-agent-03


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.7874160302267805,
    "K": 0.0010601777605335054,
    "d0": 0.0012897224766212265,
    "tau": 0.05
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L_eff": 2.4765476192620257,
    "K": 0.0009714616153136761,
    "d0": 4.480994570776638e-05,
    "tau": 0.05
  }
}
```


## m2-agent-05


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "g": 0.978386024867586,
    "K_us": 0.003952369466136279,
    "delta0": 0.0013386063106657791,
    "tau": 0.05245353568404573
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "g": 1.212362348609587,
    "K_us": 0.0030155666104823165,
    "delta0": -0.00017873648699720767,
    "tau": 0.05847378305110825
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "g": 1.0,
    "K_us": 0.003,
```

```
    "delta0": 0.0,
    "tau": 0.05
  }
}
```

m2-agent-09

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "s": 1.203330959234065,
    "K": 0.0029361363287092115,
    "bias": 0.00021448227575312877,
    "tau": 0.032147914442713796,
    "delay": 1
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "s": 0.9772602779030508,
    "K": 0.003915796445879638,
    "bias": -0.004423910419199254,
    "tau": 0.022540457646443024,
    "delay": 1
  }
}
```

m3-agent-01

```
{
  "_doc": "Per-platform fitted coefficients for the kinematic-bicy
  "tau_s": 0.08,
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.9567028992805204,
    "K_us": 0.0033801343956829,
    "delta_0": 0.0008461582112823263,
```

```
    "L": 3.7
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.1725815243702862,
    "K_us": 0.0025152837759692456,
    "delta_0": 0.00013949413303459602,
    "L": 2.984
  }
}
```


## m3-agent-03


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g0": 0.9224096029662896,
    "g2": 0.15876361430939134,
    "delta0": -0.0010913875270846236,
    "L_eff": 3.6413019931403277,
    "K0": 0.004767766094094899,
    "K1": -5.858241730532427e-05,
    "tau": 0.0740452029657968
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g0": 1.0219456519707983,
    "g2": 0.40092616819123517,
    "delta0": 5.9966690969289335e-05,
    "L_eff": 2.771534314392467,
    "K0": 0.0015320260055171069,
    "K1": 1.5004138297234027e-05,
    "tau": 0.08533887973933936
  }
}
```

m3-agent-05


```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.1935822683980741,
    "delta_offset": 0.00016378819390629683,
    "K_us": 0.0027507391842350546,
    "tau": 0.07010533877461887,
    "L": 2.984
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.9786307745986058,
    "delta_offset": -0.0012312582055424145,
    "K_us": 0.003976165863986497,
    "tau": 0.06037312785047078,
    "L": 3.7
  }
}
```


m3-agent-07


```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.1762462360941122,
    "d0": 0.00010063505427400637,
    "K_us": 0.0026382633004772204,
```

```
    "tau": 0.06840716946070041,
    "g_v3": 1.1770261504574182,
    "d0_v3": 0.00014959445374349638,
    "K_us_v3": 0.0026591289393425076,
    "tau_v3": 0.06881984429222407,
    "alpha_v3": 3.414239845943319e-09,
    "g_v1": 1.173581598798327,
    "d0_v1": 9.779075692684366e-05,
    "K_us_v1": 0.0026023048615728033,
    "L": 2.984
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.9812529513258541,
    "d0": 0.0011869609888145339,
    "K_us": 0.0041443305277957445,
    "tau": 0.06130822211279313,
    "g_v3": 0.9812464336315232,
    "d0_v3": 0.0011870345278419555,
    "K_us_v3": 0.004144503544848034,
    "tau_v3": 0.06128822752272341,
    "alpha_v3": 3.986979263377587e-09,
    "g_v1": 0.9795065634523334,
    "d0_v1": 0.0011752337517999898,
    "K_us_v1": 0.004081817473342322,
    "L": 3.7
  }
}
```

m3-agent-09

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "model": "v1_global_delta0",
    "g": 0.8633847850655261,
    "delta0": 0.001333845368641501,
    "L_eff": 3.262313536845297,
    "K_us": 0.00349772209138941,
    "tau": 0.05952805407230728
```

```
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "model": "v4_per_segment_delta0",
    "g": 0.8907859709885533,
    "L_eff": 2.2160488557450244,
    "K_us": 0.002016048675027096,
    "tau": 0.0690564359086768,
    "delta0_fallback": -0.00010084780741412298
  }
}
```


## 6b. Operating-contract audit

The grader strips `sim_df` to an allowlist before calling each agent's `predict()`, so an agent cannot read truth at inference time. This section reports source-level intent — which agents *would have* read truth columns if the grader hadn't intervened. A clean audit is its own quality signal.

> ⚠️ **4 of 29 agents** had references to truth-derived column names inside their `predict()` body. The grader's allowlist stripped these columns before calling `predict()` — so the leak cannot fire at scoring time — but the source-level intent is captured here.

**Columns attempted** (count of agents whose predict body references each):

- `a_lat_meas_mps2` — 3 agent(s)
- `yaw_rate_meas_rads` — 1 agent(s)

| agent | family | columns referenced inside predict() |
|---|---|---|
| m1-agent-06 | module-1 | yaw_rate_meas_rads |
| m3-agent-02 | module-3 | a_lat_meas_mps2 |
| m3-agent-06 | module-3 | a_lat_meas_mps2 |
| m3-agent-09 | module-3 | a_lat_meas_mps2 |

## 7. Reconstruction quality (substrate signal)

How many agents shipped the right artefacts to be canonically gradable. Failures here are a substrate or contract problem, not a model problem.

| format check | pass | fail |
| --- | --- | --- |
| agent_folder_exists | 29 | 0 |
| has_manifest_json | 29 | 0 |
| manifest_parsable | 29 | 0 |
| manifest_declares_predict_callable | 29 | 0 |
| manifest_declares_platform_support | 29 | 0 |
| has_predict_py | 29 | 0 |
| has_coeffs_json | 22 | 7 |
| has_report | 25 | 4 |

**Failure reasons across the cohort:**

- `import_failed` — 1 agent(s)

## 8. Worst-of-cohort

### Lowest CTE Δ%

- m3-agent-06 (-7.4%)

- m3-agent-02 (+11.6%)
- m2-agent-01 (+15.1%)

### Lowest yaw Δ%

- m1-agent-10 (+10.9%)
- m1-agent-06 (+10.9%)
- m1-agent-09 (+12.3%)